

Annotated Bibliography

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[3] The article provides a summary of techniques presently used in sustainable development within the scientific community. The article begins by discussing the importance of sustainable development, the frequency of various techniques used to analyze sustainable development, and important issues in sustainable transportation, such as logistics and environmental issues. It then dives into the techniques that are currently being used to analyze sustainable development, which includes simulation, optimization, machine learning, and fuzzy set analysis. The Optimization section covered the history of optimization used in transportation and the variety of applications for optimization, including in the difference between urban transportation systems and in global/regional transportation systems. Applications in simulation focus on the growing integration of simulation and optimization models and on usages in logistics, traffic congestion, and on the usage of demand-based transportation systems (ride sharing applications). The machine learning section primarily summarizes a paper that uses a random forest model to understand patterns in a bike sharing transportation system, as well as touches on a variety of techniques (random forest, logistic regression, neural networks, support vector machines, decision trees, heuristics, multi-agent systems, and multi-variate analysis) for a variety of analytical, traffic-related outcomes.

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References

- [1] Kenneth Li-Minn Ang, Jasmine Kah Phooi Seng, Ericmoore Ngharamike, and Gerald K. Ijemaru. Emerging technologies for smart cities' transportation: Geo-information, data analytics and machine learning approaches. *ISPRS International Journal of Geo-Information*, 11(2), 2022.
- [2] Yosoon Choi. Geoai: Integration of artificial intelligence, machine learning, and deep learning with gis. *Applied Sciences*, 13(6), 2023.
- [3] Rocio de la Torre, Canan G. Corlu, Javier Faulin, Bhakti S. Onggo, and Angel A. Juan. Simulation, optimization, and machine learning in sustainable transportation systems: Models and applications. *Sustainability*, 13(3), 2021.